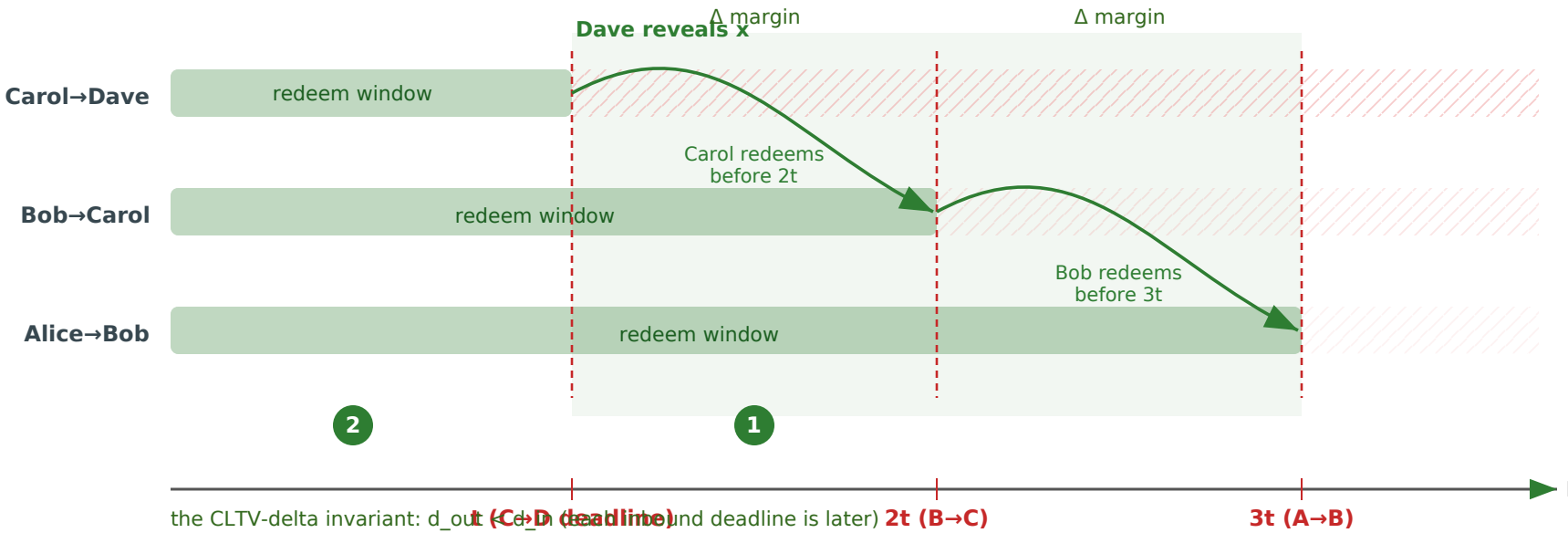
Staggered CLTV timelocks — timing safety (block height →)

covers cltv_blocks.spthy · Clock.spthy · timeout.spthy — see rail →



LEMMAS · timing theories

1. **cltv_blocks.spthy — windows EXIST**
   CLTV_Gap_Is_Positive ✓ · Claim_Window_Nonempty ✓
   Staggered_Path_Safe ✓ (derives d_out < d_in from block

2. **Clock.spthy — claim-in-time + lifecycle**
   Claim_Window_Exists ✓ · Outcome_Exclusive ✓
   Transitive_Preimage_Before_Upstream_Deadline ✓
   No_HTLC_After_Close ✓ · Redeem_Reachable / Refund_R

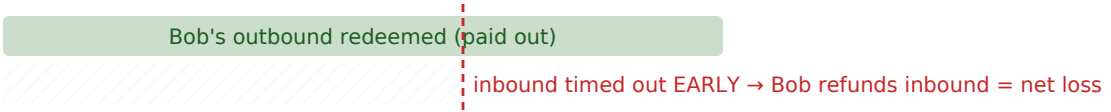
3. **timeout.spthy — the race (T2b removed)**
   Early_Timeout_Race ✓ (reachable without T2b)
   pairs with multihop's Timeout_race_Blocked ✓
   → together: proof that T2b is necessary, not assumed aw

green = timing safety proved · red = the race, shown reachable to justify the assumption.

cltv_blocks derives what multihop assumes; gaps adds the block-clock lifecycle safety.

3 WITHOUT the liveness assumption T2b (timeout.spthy)

adversary fires the A→B inbound timeout early → Bob's outbound is redeemed but his inbound refunds → Bob loses.



Two-sided certificate: race reachable here (T2b removed) ↔ blocked in multihop.spthy (Timeout_race_Blocked). So T2b is load-bearing.

Rough draft — timing theories mapped. This completes all 5 files across the 3 figures.